Solutions Tutorial 5

1. Given a function $f(x,y)$ with all continuous partial derivatives up to order three, write the first order and second order Taylor polynomials $P_1(x,y), P_2(x,y)$ near a point $P_0=(x_0,y_0)$ in terms of Δx, Δy and the values of the function and of appropriate partial derivatives at P_0.

Answer. $P_1(x,y)=f(x_0,y_0)+\Delta x f_x(x_0,y_0)+\Delta y f_y(x_0,y_0)$; and

$$P_2(x,y)=P_1(x,y)+\tfrac{1}{2}\Big[(\Delta x)^2 f_{xx}(x_0,y_0)+2(\Delta x)(\Delta y)f_{xy}(x_0,y_0)+(\Delta y)^2 f_{yy}(x_0,y_0)\Big].$$

2. Use Taylor formula to find the approximate value of $\sqrt{24}$ using linear, quadratic and cubic approximation. (Hint: Let $f(x)=\sqrt{x}$ centred at $x_0=25$.)

Solution. Naturally we let $f(x)=\sqrt{x}$ and use $x_0=25$ (so that $\sqrt{25}=5$ is close to $\sqrt{24}$). We may also use $f(x)=\sqrt{25-x}$ with $x_0=0$.

$f(x)=\sqrt{x}$	$f'(x)=\frac{1}{2}x^{-\frac{1}{2}}$	$f''(x)=\frac{(-1)^{2-1}}{2^2}x^{-\frac{2(2)-1}{2}}$	$f'''(x)=\frac{(-1)^{3-1}(2(3)-3)}{2^3}x^{-\frac{2(3)-1}{2}}$
$f(25)=5$	$f'(25)=\frac{1}{2(5)}$	$f''(25)=\frac{-1}{4(5^3)}$	$f'''(25)=\frac{3}{8(5^5)}$

So, $f^{(r)}(x)=\frac{(-1)^{r-1}(2r-3)}{2^r}x^{-\frac{2r-1}{2}} \Rightarrow f^{(r)}(25)=\frac{(-1)^{r-1}(2r-3)}{2^r 5^{2r-1}}$. Therefore,

$$f(x)=\sum_{r=0}^{\infty}\frac{f^{(r)}(25)}{r!}(x-25)^r=5+\frac{x-25}{2(5)}+\sum_{r=2}^{\infty}\frac{(-1)^{r-1}(2r-3)}{2^r 5^{2r-1} r!}(x-25)^r.$$

Approximation	Taylor's formula	Approximation
Linear	$p_1(x)=5+\frac{x-25}{2\times 5}$	$f(24)\approx p_1(24)=5+\frac{-1}{10}=4.9$
Quadratic	$p_2(x)=p_1(x)-\frac{(x-25)^2}{4\times 5^3\times 2!}$	$f(24)\approx p_2(24)=4.9-\frac{1}{1000}=4.899$
Cubic	$p_3(x)=p_2(x)+\frac{3(x-25)^3}{8\times 5^5\times 3!}$	$f(24)\approx p_3(24)=4.899-\frac{3}{150000}=4.89898$

3. Use Taylor formula to find the approximate value of $e=2.71828...$ using linear, quadratic, cubic and quartic approximation

Solution. $f^{(r)}(x)=e^x \Rightarrow f^{(r)}(0)=e^0=1$. So, $f(x)=e^x=\sum_{r=0}^{\infty}\frac{x^r}{r!}$ and $f(1)=e^1=\sum_{r=0}^{\infty}\frac{1}{r!}$.

$e^1\approx\frac{1}{0!}=1$	
$e^1\approx\frac{1}{0!}+\frac{1}{1!}=1+1=2$	(Linear approximation)
$e^1\approx\frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}=2+\frac{1}{2}=2.5$	(Quadratic approximation)
$e^1\approx\frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}=2.66666...$	(Cubic approximation)
$e^1\approx\frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}=2.708333...$	(Quartic approximation)

4. Find the Taylor expansions of the following functions about the given point up to the second order:

(a) $f(x,y) = e^x \cos y$ at $(0, 0)$ (b) $f(x,y) = \sin(xy)$ at $(1, \frac{\pi}{2})$

(c) $f(x,y) = \frac{y^2}{x^3}$ at $(1, -1)$ (d) $f(x,y) = e^{x+y}$ at $(2, -2)$

Solution. $P_1(x,y) = f(x_0,y_0) + \Delta x f_x(x_0,y_0) + \Delta y f_y(x_0,y_0)$;

$$P_2(x,y) = P_1(x,y) + \tfrac{1}{2}\Big[(\Delta x)^2 f_{xx}(x_0,y_0) + 2(\Delta x)(\Delta y) f_{xy}(x_0,y_0) + (\Delta y)^2 f_{yy}(x_0,y_0)\Big].$$

	(a) at $(0, 0)$	(b) at $(1, \frac{\pi}{2})$	(c) at $(1, -1)$	(d) at $(2, -2)$
$f(x,y)$	$e^x \cos y$	$\sin(xy)$	$y^2 x^{-3}$	$e^{x+y} = e^x e^y$
$f(a,b)$	1	1	1	1
$f_x(x,y)$	$e^x \cos y$	$y\cos(xy)$	$-3y^2 x^{-4}$	$e^x e^y$
$f_x(a,b)$	1	0	–3	1
$f_y(x,y)$	$-e^x \sin y$	$x\cos(xy)$	$2yx^{-3}$	$e^x e^y$
$f_y(a,b)$	0	0	–2	1
$f_{xx}(x,y)$	$e^x \cos y$	$-y^2 \sin(xy)$	$12y^2 x^{-5}$	$e^x e^y$
$f_{xx}(a,b)$	1	$-\frac{\pi^2}{4}$	12	1
$f_{xy}(x,y)$	$-e^x \sin y$	$\cos(xy) - xy \sin(xy)$	$-6yx^{-4}$	$e^x e^y$
$f_{xy}(a,b)$	0	$-\frac{\pi}{2}$	6	1
$f_{yy}(x,y)$	$-e^x \cos y$	$-x^2 \sin(xy)$	$2x^{-3}$	$e^x e^y$
$f_{yy}(a,b)$	–1	–1	2	1

(a) $P_1(x,y) = 1 + x(1) + y(0) = 1 + x$;

$$P_2(x,y) = P_1(x,y) + \tfrac{1}{2}\Big[x^2(1) + 0 + y^2(-1)\Big] = 1 + x + \tfrac{1}{2}(x^2 - y^2).$$

(b) $P_1(x,y) = 1$;

$$P_2(x,y) = 1 + \tfrac{1}{2}\Big[(x-1)^2(-\tfrac{\pi^2}{4}) + 2(x-1)(y-\tfrac{\pi}{2})(-\tfrac{\pi}{2}) + (y-\tfrac{\pi}{2})^2(-1)\Big]$$
$$= 1 - \tfrac{\pi^2}{8}(x-1)^2 - \tfrac{\pi}{2}(x-1)(y-\tfrac{\pi}{2}) - \tfrac{1}{2}(y-\tfrac{\pi}{2})^2.$$

(c) $P_1(x,y) = 1 - 3(x-1) - 2(y+1)$;

$$P_2(x,y) = 1 - 3(x-1) - 2(y+1) + \tfrac{1}{2}\Big[12(x-1)^2 + 12(x-1)(y+1) + 2(y+1)^2\Big]$$
$$= 1 - 3(x-1) - 2(y+1) + 6(x-1)^2 + 6(x-1)(y+1) + (y+1)^2.$$

(d) $P_1(x,y) = 1 + (x-2) + (y+2)$;

$$P_2(x,y) = 1 + (x-2) + (y+2) + \tfrac{1}{2}(x-2)^2 + (x-2)(y+2) + \tfrac{1}{2}(y+2)^2$$

5. Estimate the error $R_n(x)$ in the Taylor expansion at x_0 for the following functions.

(a) $f(x)=\cos(\frac{x}{3})$ at $x_0=0$ estimate $R_3(x)$ on the interval $\left|x\right|\le\frac{1}{4}$.

(b) $f(x)=e^x$, $x_0=0$ estimate $R_3(x)$ on the interval $\left|x\right|\le\frac{1}{2}$.

(c) $f(x)=\sqrt{1+x}$, $x_0=0$ estimate $R_2(x)$ on the interval $\left|x-1\right|\le 1$.

(d) $f(x)=x^{1/3}$, $x_0=1$ estimate $R_3(x)$ on the interval $1\le x\le 2$.

Solution. $R_{n+1}(x)=\frac{(x-x_0)^{n+1}}{(n+1)!}f^{(n+1)}(\xi)$, ξ is in between x and x_0.

(a) $f^{(3)}(x)=\frac{1}{27}\sin(\frac{x}{3})$. Given $\left|x\right|\le\frac{1}{4}$, then

$$\left|R_3(x)\right|=\left|\frac{x^3}{3!}\left(\frac{1}{27}\sin\left(\frac{\xi}{3}\right)\right)\right|\le\frac{\left|x\right|^3}{27(6)}\le\frac{1}{4^3(27)(6)}=\frac{1}{10368}.$$

(b) $f^{(3)}(x)=e^x$. Given $\left|x\right|\le\frac{1}{2}$ and ξ is in between x and 0 (and hence $-\frac{1}{2}\le\xi\le\frac{1}{2}$ because x is in between $\pm\frac{1}{2}$), and with the fact that $e<3$, so we have

$$\left|R_3(x)\right|=\left|\frac{x^3}{3!}e^{\xi}\right|\le\frac{e^{\xi}}{2^3(6)}\le\frac{e^{0.5}}{48}\le\frac{\sqrt{3}}{48}<\frac{2}{48}=\frac{1}{24}.$$

*If we know the value of $\sqrt{e}$ then we could substitute the value and obtain a more precise error. **If we do not know the exact answer of $\sqrt{3}$, we may take $\sqrt{3}<2$ because $1<3<4\Rightarrow 1<\sqrt{3}<2$.

(c) $f''(x)=\frac{-1}{4\sqrt{(1+x)^3}}$. Given $\left|x-1\right|\le 1$ (or $0\le x\le 2$) and $0\le\xi\le 2$, so

$$\left|R_2(x)\right|=\frac{x^2}{2!4\sqrt{(1+\xi)^3}}\le\frac{4}{2(4)\sqrt{(1+\xi)^3}}\le\frac{4}{2(4)\sqrt{(1+0)^3}}=\frac{1}{2}.$$

(d) $f'''(x)=\frac{10}{27}x^{-8/3}$. $1\le x\le 2\Rightarrow 0\le x-1\le 1$ so $(x-1)^3\le 1$ and $1\le\xi\le 2$, so

$$\left|R_3(x)\right|=\left|\frac{10(x-1)^3}{3!27\xi^{8/3}}\right|\le\left|\frac{10(1)^3}{3!27(1^{8/3})}\right|\le\frac{5}{81}.$$

Extra Exercises

6. Find the Taylor cubic approximation to the function $f(x) = \sin x$ centred at $x = 0$; and draw its graph. Do the same for the quintic approximation. Compare both approximations with the graph of $\sin x$.

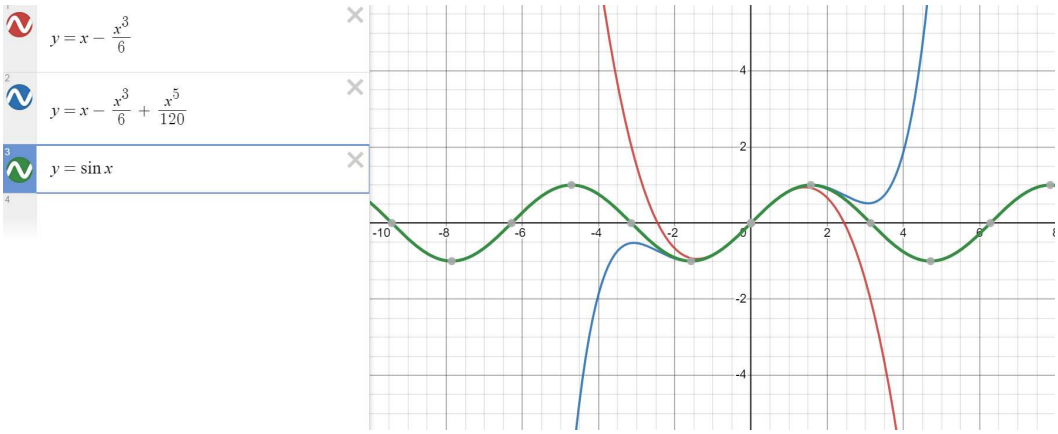
Solution. Note that $f(x) = \sin x = f^{(4n)}(x) \Rightarrow f^{(4r)}(0) = 0$

$f'(x) = \cos x = f^{(4n+1)}(x) \Rightarrow f^{(4r+1)}(0) = 1$

$f''(x) = -\sin x = f^{(4n+2)}(x) \Rightarrow f^{(4r+2)}(0) = 0$

$f'''(x) = -\cos x = f^{(4n+3)}(x) \Rightarrow f^{(4r+3)}(0) = -1$

So, $f(x) = \sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \ldots = \sum_{r=1}^{\infty} \frac{(-1)^{r-1}x^{2r-1}}{(2r-1)!}$. The cubic and quintic approximations are $\sin x \approx x - \frac{x^3}{3!}$ and $\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!}$, respectively.



7. Find the Taylor approximation to order n of $f(x) = e^{-x}$ centred at $x = -4$.

Solution. $f(x) = e^{-x} \Rightarrow f^{(r)}(x) = (-1)^r e^{-x} \Rightarrow f^{(r)}(-4) = (-1)^r e^4$.

So, the Taylor approximation to order n of $f(x) = e^{-x}$ centred at $x = -4$ is

$$f(x) = e^{-x} \approx \sum_{r=0}^{n} \frac{(-1)^r e^4}{r!}(x+4)^r = e^4 \sum_{r=0}^{n} \frac{(-1)^r (x+4)^r}{r!}.$$

8. Prove the following by finding the Taylor polynomial of order n, centred at $x = 0$:

(a) $\frac{1}{1+x} \approx \sum_{k=0}^{n} (-1)^k x^k$ (b) $\log(x+1) = \ln(x+1) \approx \sum_{k=1}^{n} \frac{(-1)^{k+1}x^k}{k}$

Proof. (a) $f(x) = (1+x)^{-1} \Rightarrow f^{(k)}(x) = (-1)^k k!(1+x)^{-k-1} \Rightarrow f^{(k)}(0) = (-1)^k k!$.

$$\therefore \frac{1}{1+x} \approx \sum_{k=0}^{n} \frac{(-1)^k k!}{k!} x^k = \sum_{k=0}^{n} (-1)^k x^k.$$

(b) $g(x) = \ln(1+x) \Rightarrow g^{(k)}(x) = f^{(k-1)}(x) \Rightarrow g^{(k)}(0) = f^{(k-1)}(x) = (-1)^{k-1}(k-1)!, k \geq 1$

$$\ln(x+1) \approx g(0) + \sum_{k=1}^{n} \frac{(-1)^{k-1}(k-1)!}{k!} x^k = 0 + \sum_{k=1}^{n} \frac{(-1)^{k+1}x^k}{k} = \sum_{k=1}^{n} \frac{(-1)^{k+1}x^k}{k}.$$

Note that $(-1)^{k-1} = (-1)^{k+1}\ \forall k \in \mathbb{N}$.

9. Obtain the Taylor polynomial of order n for $f(x)=\log x$ about $x=2$.

Solution. $f(x)=\log x \Rightarrow f^{(k)}(x)=\frac{(-1)^{k-1}(k-1)!}{x^k} \Rightarrow f^{(k)}(2)=\frac{(-1)^{k-1}(k-1)!}{2^k}, k\geq 1$.

$$f(x)=\log x \approx f(2)+\sum_{k=1}^{n}\frac{(-1)^{k-1}(k-1)!2^{-k}}{k!}(x-2)^k=\log 2+\sum_{k=1}^{n}\frac{(-1)^{k-1}}{k2^k}(x-2)^k .$$

10. Prove that $\sin x=\sum_{k=1}^{n}\frac{(-1)^{k-1}x^{2k-1}}{(2k-1)!}+E_{2n}(x)$, $\left|E_{2n}(x)\right|\leq\frac{|x|^{2n+1}}{(2n+1)!}$; and

$$\cos x=\sum_{k=0}^{n}\frac{(-1)^{k}x^{2k}}{(2k)!}+E_{2n+1}(x), \left|E_{2n+1}(x)\right|\leq\frac{|x|^{2n+2}}{(2n+2)!}.$$

Proof. $R_{n+1}(x)=\frac{(x-x_0)^{n+1}}{(n+1)!}f^{(n+1)}(\xi)$, ξ is in between x and x_0. Here, $x_0=0$, we have $R_{n+1}(x)=\frac{x^{n+1}}{(n+1)!}f^{(n+1)}(\xi)$, ξ is in between x and 0. Also note that when the function

$$f(x)=\sin x \Rightarrow f^{(n)}(x)=\begin{cases}\sin x, & n\equiv 0 \pmod 4\\ \cos x, & n\equiv 1 \pmod 4\\ -\sin x, & n\equiv 2 \pmod 4\\ -\cos x, & n\equiv 3 \pmod 4\end{cases}\Rightarrow\left|f^{(n)}(\xi)\right|\leq 1,\ \xi \text{ is between } x \text{ and } 0.$$

(a) From Q6: $\sin x=\sum_{k=1}^{n}\frac{(-1)^{k-1}x^{2k-1}}{(2k-1)!}+E_{2n}$, where

$$\left|E_{2n}(x)\right|=\left|\frac{x^{2n+1}}{(2n+1)!}f^{(2n+1)}(\xi)\right|\leq\frac{|x|^{2n+1}}{(2n+1)!} \text{ since } \left|f^{(2n+1)}(\xi)\right|\leq 1.$$

(b) $\cos x=\sum_{k=0}^{n}\frac{(-1)^{k}}{(2k)!}x^{2k}+E_{2n+1}(x)$, where

$$\left|E_{2n+1}(x)\right|=\left|\frac{x^{2n+2}}{(2n+2)!}f^{(2n+2)}(\xi)\right|\leq\frac{|x|^{2n+2}}{(2n+2)!} \text{ since } \left|f^{(2n+2)}(\xi)\right|\leq 1.$$

11. Find an approximate value for $f(x,y)=\sin(\pi xy+\ln y)$ at $(0.01,\ 1.05)$.

Solution.
$$\begin{aligned} f(x,y)&=\sin(\pi xy+\ln y) && \Rightarrow f(0,1)=0\\ f_x&=\pi y\cos(\pi xy+\ln y) && \Rightarrow f_x(0,1)=\pi\\ f_y&=\left(\pi x+\tfrac{1}{y}\right)\cos(\pi xy+\ln y) && \Rightarrow f_y(0,1)=1\end{aligned}$$

So, $f(0.01,1.05)\approx f(0,1)+0.01f_x(0,1)+0.05f_y(0,1)=0.01\pi+0.05\approx 0.0814$.

12. If two resistors of resistance R_1 and R_2 are wired in parallel, then the resulting resistance R satisfies the equation $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}$. Use the linear approximation to estimate the change in R if R_1 decreases from 2 to 1.9 ohms and R_2 increases from 8 to 8.1 ohms.

Solution. $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}\Rightarrow -\frac{1}{R^2}\frac{\partial R}{\partial R_i}=-\frac{1}{R_i^2}\Rightarrow \frac{\partial R}{\partial R_i}=\frac{R^2}{R_i^2}, i=1,2$. When $(R_1,R_2)=(2,8)$,

$R(2,8)=\frac{1}{\frac{1}{2}+\frac{1}{8}}=\frac{8}{5}$. So, $\left.\frac{\partial R}{\partial R_i}\right|_{(2,8)}=\begin{cases}\left(\frac{8}{5}\times\frac{1}{2}\right)^2, i=1\\ \left(\frac{8}{5}\times\frac{1}{8}\right)^2, i=2\end{cases}=\begin{cases}\frac{16}{25}, i=1\\ \frac{1}{25}, i=2\end{cases}$.

So, $R(1.9,8.1)\approx R(2,8)+\frac{16}{25}(-0.1)+\frac{1}{25}(0.1)=-0.06$.

13. Let $f(x,y)=\frac{2y}{x^2+y^2}$. Find an approximate value for $f(-0.8,2.1)$.

Solution. $f(x,y)=\frac{2y}{x^2+y^2}\Rightarrow f(x_0,y_0)=f(-1,2)=\frac{4}{5}$

$f_x=-\frac{4xy}{(x^2+y^2)^2}\Rightarrow f_x(-1,2)=\frac{8}{25}\qquad f_y=\frac{2}{x^2+y^2}-\frac{4y}{(x^2+y^2)^2}\Rightarrow f_y(-1,2)=-\frac{6}{25}$

$f(-0.8,2.1)\approx f(-1,2)+0.2f_x(-1,2)+0.1f_y(-1,2)=\frac{4}{5}+0.2\times\frac{8}{25}-0.1\times\frac{6}{25}=0.84.$

14. A surface is defined implicitly by $z^4-xy^2z^2+y=0$.

(a) Compute z_x and z_y in terms of x, y and z; and evaluate them at $(2,-\frac{1}{2},1)$.

(b) If x decreases from 2 to 1.94, and y increases from –0.5 to –0.4, find the approximate change in z from 1.

(c) Find the equation of the tangent plane to the surface at the point $(2,-\frac{1}{2},1)$.

Solution. (a) Note that $(2,-\frac{1}{2},1)$ satisfies the equation $z^4-xy^2z^2+y=0$. Now partially (and implicitly) differentiate the equation with respect to x, and y to obtain $z_x=\frac{(yz)^2}{4z^3-2xy^2z}$ and $z_y=\frac{2xyz^2-1}{4z^3-2xy^2z}$. So, $z_x(2,-\frac{1}{2})=\frac{1}{12}$, and $z_y(2,-\frac{1}{2})=-1$.

(b) $dz=z_xdx+z_ydy=\frac{1}{12}(-0.06)-1(0.1)=-0.105$.

(c) Tangent plane equation: $z=1+\frac{1}{12}(x-2)-(y+\frac{1}{2})\Rightarrow \frac{x}{12}-y-z=-\frac{1}{3}$.

Past years test/final exam questions

15. Consider the function $f(x,y) = e^{-x^2+4y^2}$.

(a) Find the equation of the tangent plane to the graph z at the point where (2, 1).

(b) Find the tangent plane approximation to the value of $f(1.99,\ 1.01)$ using the tangent plane from part (a).

Solution. (a) $f(x,y) = e^{-x^2+4y^2} \Rightarrow f_x = -2xe^{-x^2+4y^2}, f_y = 8ye^{-x^2+4y^2}$. So the tangent plane is $z = f(2,1) + (x-2)f_x(2,1) + (y-1)f_y(2,1) = e^0 - 4(x-2) + 8(y-1) = 1 - 4x + 8y$.

(b) Since $f(1.99,1.01) \approx f(2,1) + (x-2)f_x(2,1) + (y-1)f_y(2,1) = 1 - 4x + 8y$, it follows that $f\big(1.99,1.01\big) \approx 1 - 4(1.99-2) + 8(1.01-1) = 1.12$.

16. Find a linear approximation value for $\frac{0.99^2}{1.001} + 1.01^3$. Give your answer correct to three decimal places.

Solution. Let $f(x,y,z) = \frac{x^2}{y} + z^3$. Then $f_x = \frac{2x}{y}$, $f_y = -\frac{x^2}{y^2}$, $f_z = 3z^2$.

Let $x = 0.99, y = 1.001, z = 1.01$, $x_0 = 1 = y_0 = z_0$.

Then $\Delta x = -0.01, \Delta y = 0.001, \Delta z = 0.01$. Thus,

$f(x,y,z) \approx f(x_0,y_0,z_0) + (\Delta x)f_x(x_0,y_0,z_0) + (\Delta y)f_y(x_0,y_0,z_0) + (\Delta z)f_z(x_0,y_0,z_0)$

$\Rightarrow \frac{0.99^2}{1.001} + 1.01^3 \approx 2 + (-0.01)(2) + (0.001)(-1) + (0.01)(3) = 2.009$.